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Selective degradation of hemicellulose and lignin for improving enzymolysis efficiency via pretreatment using deep eutectic solvents

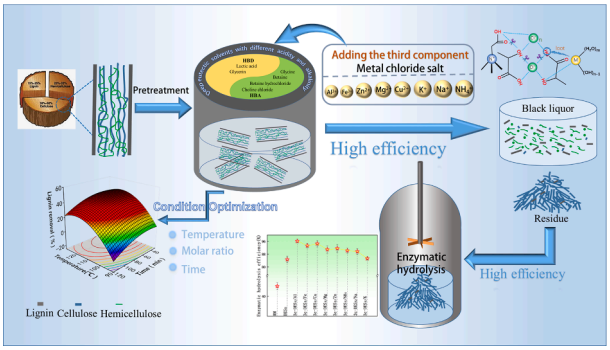
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HIGHLIGHTS

- Deep eutectic solvents pretreatment was sustainability and low carbon.
- Carboxyl, Cl^- and N^+ in DESs had stronger hydrogen bond destruction ability.
- DESs-$AlCl_3$ had excellent efficiency for its polyhedral coordination species.
- 90% of non-cellulosic constituent removal improved the enzymolysis efficiency.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:
Lignocellulosic biomass
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Metal chloride salt
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ABSTRACT

Deep eutectic solvents (DESs) with different acidity and alkalinity were applied for biomass pretreatment, and the conditions were optimized by response surface methodology. The results showed that lactic acid/betaine hydrochloride had the optimal pretreatment efficiency, where the removal rates of hemicellulose and lignin came up to 89% and 73%, and the enzymolysis efficiency was as high as 92%. Furthermore, eight types of chloride salts with different valence states were introduced into the DESs as the third component. The chloride salts could improve the pretreatment efficiency and positively correlated with the metal valence state. Specifically, $AlCl_3$ was significantly superior in improving the pretreatment efficiency, where the enzymolysis efficiency reached 96% due to the destruction of crystalline region and the esterification of partial cellulose. Therefore, it is proposed that adding highly valent metal salts to acidic DESs has higher pretreatment and enzymatic efficiency.

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1. Introduction

The depletion of unrenewable resources and the conjuncture of ecological circumstances lead to an increasing interest in lignocellulosic biomass as a green, renewable, and sustainable resource (Yiin et al., 2021). However, the compact structure and complex chemical composition of the lignocellulosic biomass greatly hinder its high-valued utilization. (Ma et al., 2015). Therefore, pretreatment is essential to improve lignocellulosic biomass utilization by disrupting the physicochemical structure and selectively separating components (Mankar et al., 2021a).

Recently, deep eutectic solvents (DESs) have been regarded as effective green solvents that can effectively dissolve and separate the components of lignocellulosic biomass. Abbott and his colleagues defined the DESs as low-melting-temperature mixtures, and they synthesize a transparent solution using choline chloride and urea at low temperatures. (Abbott et al., 2004). Compared with Ionic Liquids, DESs have the advantages of easy preparation, low price, less toxic, green and easy recycling, etc., which makes it useful in many fields (Carriazo et al., 2012; Wang et al., 2018). Currently, the acidic DESs are primarily investigated, mainly consisting of organic acids such as lactic acid, malonic acid, and citric acid as hydrogen-bond donors (HBD), combined with the hydrogen-bond acceptor (HBA). The acidic DESs are suitable for the pretreatment of various lignocellulosic materials. When the walnut, peach endocarp and coconut are pretreated with the DES composed of lactic acid/choline chloride, the delignification rate can reach 66.3%, 70.2% and 82%, respectively, with the purity above 90% (Li et al., 2017; Li et al., 2018; Mankar et al., 2021b). In addition to acid-based DESs, alcohol-based DESs have also been concerned. Zhang et al. (Zhang et al., 2016) pretreated corn cobs using ethylene glycol/ChCl and glycerol/ChCl at the same molar ratio condition (2:1), and delignification rates were up to 87.6% and 71.3%, respectively. Further studies have found that the separation efficiency of the lignocellulosic biomass drastically increased by adding a third component into the binary DESs. (Lee et al., 2009;Xia et al., 2018). Generally, the third component has the following effects: reducing the viscosity of the DESs (Kandanelli et al., 2018;Kumar et al., 2016), enhancing the lignin solubility in the DESs system, and providing active protons and acidic sites (Chen et al., 2018). According to the previous research, metal chlorides can selectively catalyze the hemicellulose degradation from lignocellulosic biomass, exhibiting diverse effects with different valence states of the metal cations (Kamireddy et al., 2013). Based on the above analysis, metal cations are introduced into the DESs to increase the acidic sites and promote the separation efficiency of the lignocellulosic biomass.

In the present work, the representative acid (lactic acid) and alcohol (glycerol) are utilized as the HBD to combine with four types of HBA for DESs preparation. Then the DES with the best separation effect is optimized and investigated in the pretreatment of eucalyptus wood chips by response surface methodology. Furthermore, metal chloride salts with different valence states are added into the DESs, and the variation of the H-bond ability was analyzed by observing the polarity parameters. The purpose of this research is to clarify the mechanism of the selective action of binary and ternary DESs on the lignocellulosic biomass and establish a theoretical basis for the efficient and environment-friendly conversion of lignocellulosic biomass.

2. Materials and methods

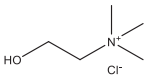
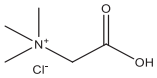
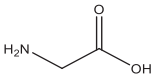
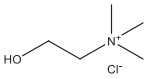
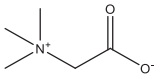
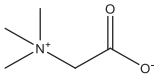
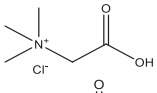
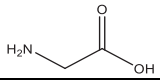
2.1. Materials

Eucalyptus wood chips were supplied by Guangxi Jingui Paper Co., Ltd (Guigang, China). Prior to using, the wood chips were washed and air-dried, then ground into powders and sieved, where the fraction of 40–60 meshes were selected. Other chemicals were analytical grade, purchased from Sinopharm Chemical Reagent Co., Ltd. (Shanghai, China).

2.2. Preparation of deep eutectic solvents

For the lactic acid-based DESs: the lactic acid was mixed with choline chloride, betaine, betaine hydrochloride and glycine under 80 °C to form clear solutions, and correspondingly they were labeled as 1–1, 1–2, 1–3

Table 1
Kamlet–Taft parameters for different Deep Eutectic Solvents.

No.	Deep Eutectic Solvents	HBD[a]	HBA[b]	Chemical structural formula of HBA	Molar ratio HBD: HBA	Kamlet–Taft parameters		
						π*	α	β
1	1–1	Lactic acid	Choline chloride		2:1	1.00	1.34	0.51
2	1–2		Betaine		2:1	1.02	1.30	0.54
3	1–3		Betaine hydrochloride		6:1	0.96	2.09	0.65
4	1–4		Glycine		7:1	1.06	1.22	0.55
5	2–1	Glycerin	Choline chloride		2:1	1.08	0.86	0.65
6	2–2		Betaine		2:1	1.06	0.97	0.62
7	2–3		Betaine hydrochloride		10:1	1.06	0.93	0.60
8	2–4		Glycine		17:1	0.98	0.90	0.66

[a] HBD denotes hydrogen-bond donors.
[b] HBA denotes hydrogen-bond acceptor.

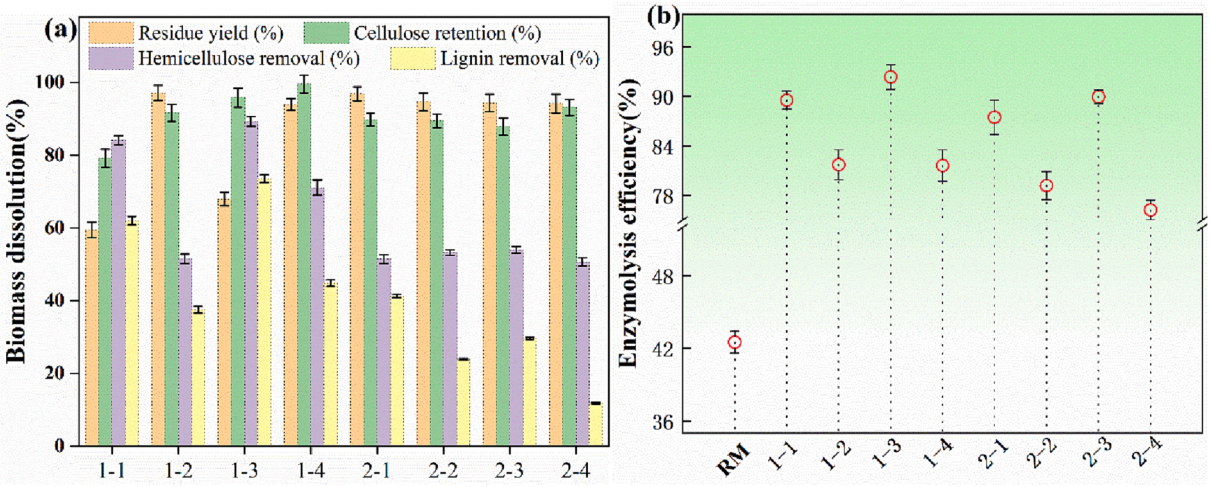


Fig. 1. Effects of the deep eutectic solvents on the pretreatment and followed enzymolysis: (a) Components changes of the solid residues (b) Enzymolysis efficiency of the solid residues.

and 1–4. For the glycerol-based DESs: the glycerol was also mixed with the choline chloride, betaine, betaine hydrochloride and glycine under the same conditions, and sequentially they were labeled as 2–1, 2–2, 2–3 and 2–4.

2.3. Pretreatment of deep eutectic solvents

The eucalyptus wood powders and DESs were mixed with a mass ratio of 1:10 and held strictly at 120 °C for 3 h in a flask and terminated by adding ethanol. The mixture was filtered with a vacuum. The solid residues were repeatedly washed with ethanol/water (90:10 v/v) and deionized water until the filtrate was clarified and neutral, air-dried.

2.4. Optimization analysis with response surface methodology

The three main factors that affected the pretreatment efficiency (Qiu et al., 2017), including pretreatment temperature (A), pretreatment time (B) and the molar ratio of DESs (C), were investigated by the Box-Behnken design (BBD) method. Thus, a three-factor three-level Box–Behnken design was used to optimize the pretreatment, as shown in Table 1.

2.5. Preparation and pretreatment of three-constituent deep eutectic solvents

Lactic acid, betaine hydrochloride and metal chloride salts including $FeCl_3{\cdot}6H_2O$, $AlCl_3$, $CuCl_2{\cdot}2H_2O$, KCl, NaCl, $MgCl_2$, NH_4Cl and $ZnCl_2$ were mixed at 80 °C with a certain molar ratio to prepare the three-constituent deep eutectic solvent (3c-DESs), marked as DESs/$FeCl_3$, DESs/$AlCl_3$, DESs/$CuCl_2$, DESs/$ZnCl_2$, DESs/$MgCl_2$, DESs/NaCl, DESs/NH_4Cl, and DESs/KCl, respectively. The eucalyptus wood powders and 3c-DESs were mixed and reacted at 90–130 °C for 30–120 min, with a fixed mass ratio of 1:10. After reaction, the solid residues were sequentially washed using ethanol/water and deionized water, and the conditions were as the same as the above (Xia et al., 2018).

2.6. Determination of Kamlet–Taft parameters

The determination of Kamlet–Taft (K-T) parameters has been described in detail in previous research (Xia et al., 2018), where N, N-dimethyl-4-nitroaniline is applied as the indicator for π^*, Nile red dye and 4-nitroaniline are the indicators for α and β, respectively.

2.7. Enzymolysis process

Enzymatic hydrolysis was performed in duplicate according to the Laboratory Analytical Procedure of NREL (Kim et al., 2019). 1 g of the solid residues were added into the buffer solution composed of acetic acid and sodium acetate (pH 4.8, 0.1 mol/L), then added 30 FPU/g cellulase, 37.25 CBU/g cellobiase and three drops of ethyl acetate. The above mixture was placed into a shaker at 50 °C for 72 h, with a rotational speed of 150 rpm.

2.8. Characterization of the raw material and solid residues

The hemicellulose, cellulose and lignin contents in the solid residues were determined by the NREL LAB method, where a two-step hydrolysis using H_2SO_4 was conducted (Kim et al., 2019). High-performance liquid chromatography (HPLC, Agilent 2000) equipped with a Bio-Rad Aminex HPX-87H (300×7.8 mm) column was used to determine the sugars contents, and the determination conditions were as follows: column temperature 45 °C, mobile phase H_2SO_4 (0.005 mol/L) with a flow rate of 0.6 mL/min; injection volume 10 μL, and RID detector. The acid-soluble lignin in the hydrolysis solution was determined by spectrophotometry at 205 nm, and the unhydrolyzed part was used to measure the acid-insoluble lignin. Sugars and by-products in pretreatment liquors were also analyzed by HPLC after the removal of undissolved cellulose solids and dissolved lignin fractions.

The raw material and the solid residues from the DESs pretreatment were characterized by Fourier transform infrared (FTIR, Tensor II, Bruckner, Germany). Approximately 1% finely ground samples were mixed with KBr to press a disk for detecting, and the wavenumber range and spectral resolution were 500–4000 cm^{-1} and 2 cm^{-1}, respectively.

A scanning electron microscope (SEM, JSMIT300LV, JEOL, Japan) was used to observe the microscopic structure of the solid materials. Prior to observing, the samples were fixed on the conductive tape and sprayed with gold powder.

The zeta potential of the residue slurry was analyzed using a zeta potential analyzer (FPA, German AFG Company)), where the solid residues were ultrasonically dispersed separately, and diluted into suspensions with deionized water (mass fraction 0.1%), and the scanning temperature and time were 25 °C and 30 s, respectively. The average value was taken three times.

For the X-ray diffraction (XRD) analysis, the freeze-dried samples were detected using an X-ray diffractometer (Bruker D8 Advance, Bruker Co., Germany), with Ni-filtered Cu Kα radiation at 40 Kv and 40 mA. The range of scattering angle (2θ) was from 10° to 30°, with a scan

Table 2
Optimization of the pretreatment process and enzymolysis using the response surface methodology.

No.	Factors			Hemicellulose removal rate,%	Cellulose retention,%	Lignin removal rate, %	Solids yield,%	Cellulase hydrolysis efficiency,%
	A[a]	B*b*	C*c*					
1	90	75	6	43±0.21	93±1.87	33±0.13	79±1.71	75±1.15
2	110	120	6	93±1.10	89±1.33	49±0.22	60±1.53	95±1.94
3	110	30	6	87±0.62	90±1.51	47±0.19	66±1.32	84±1.32
4	130	75	6	96±0.53	80±1.15	64±0.35	50±0.91	83±1.64
5	90	30	10.5	39±0.11	94±2.14	11±0.11	90±2.12	59±0.71
6	90	120	10.5	47±0.15	91±1.62	21±0.10	89±1.64	72±1.32
7	110	75	10.5	86±1.41	89±2.21	71±0.81	70±1.21	71±1.43
8	110	75	10.5	84±0.92	90±1.90	66±0.41	72±1.72	68±1.54
9	110	75	10.5	85±1.23	89±1.45	63±0.50	70±1.53	67±1.17
10	110	75	10.5	93±0.64	86±1.22	72±0.62	63±1.32	69±0.98
11	110	75	10.5	84±0.86	90±1.57	60±0.71	70±1.51	71±1.25
12	130	30	10.5	91±0.97	91±1.48	56±0.33	60±1.28	83±1.54
13	130	120	10.5	97±0.62	72±0.78	77±0.66	63±1.71	88±1.73
14	90	75	15	43±0.09	85±1.83	22±0.14	81±1.44	77±1.60
15	110	30	15	68±0.34	87±2.14	50±0.21	80±1.68	76±0.84
16	110	120	15	87±1.31	89±1.95	64±0.39	67±1.23	82±1.70
17	130	75	15	98±0.69	54±0.46	62±0.24	69±1.11	82±1.36

[a] A denotes pretreatment temperature, B denotes pretreatment time, and C denotes the molar ratio of DESs.

speed of 2°/min. The crystallinity index (CrI) was calculated according to the following equation (Wang et al., 2020):

$$CrI = \frac{(I_{002}\text{-}I_{am})}{I_{002}} \quad (1)$$

Where I_{002} is the maximum intensity of the (002) lattice diffraction; I_{am} is the intensity diffraction at $2\theta = 18°$.

3. Results and discussion

3.1. Determination of Kamlet-Taft parameters

The polarity of DESs can be quantified by the Kamlet-Taft parameters, where α, β and π^* represented the H-bond providing capacity, H-bond accepting capacity and polarity/polarizability, respectively. The α and β were critical for the pretreatment efficiency, where a higher α value indicated that more H^+ would be released, while the higher β value represented more H-bonds would be formed (Florindo et al., 2018). Therefore, the pretreatment efficiency of DESs could be inferred in advance by quantifying the difficulty of H-bond formation. It was indicated in Table 1 that the α value of lactic acid-based DESs was higher than that of glycerol-based DESs. Fig. 1a illustrated that the removal efficiency of the lignin and hemicellulose in the pretreatment using lactate-based DESs was much higher than that of glycerol-based DESs, which was confirmed by the previous research (Florindo et al., 2018), mainly because that lactic acid contained more carboxylic acid groups, which had a more substantial H-bond capacity to break the weaker H-bond connecting the glycosidic bonds in hemicellulose, and the β-O-4 bonds in lignin. On the other hand, the *p*Ka value of HBD in DESs might also play an essential role in delignification. Apparently, the lactic acid-based DESs with higher acidity was more suitable for the delignification, which also confirmed by Mankar et al. (Mankar et al., 2021b), indicating that the highly acidic HBD could improve the dissolution of the acid-soluble lignin in DESs.

It was evident from Table 1 that the DESs consisting of lactic acid and betaine hydrochloride (1–3) had higher values of the α (2.09) and β (0.65), which was because the betaine hydrochloride contained not only chloride and nitrogen ions but also carboxylic acid groups, making it had strong electron accepting and donating capacities (Pandey et al., 2014). Factually, the DESs 1–3 indeed had the best pretreatment efficiency, as shown in Fig. 1a. Therefore, the Kamlet-Taft parameter of DESs could be applied to estimate its pretreatment effect on lignocellulose.

3.2. Effects of the pretreatment on lignocellulose separation and enzymolysis

As illustrated in Fig. 1(a), the glycerol-based DESs had a poor effect on removing hemicellulose and lignin from lignocellulose, similar to the results of Hou et al. (2018). The more hydroxyl groups in glycerol increased the viscosity of DESs and thus negatively affected the dissolution of lignin and hemicellulose (Xia et al., 2018).

In addition, the carboxylic acid group in lactic acid could provide more protons to crack the glycoside, ester, and ether bonds than that of the hydroxyl group in glycerol, resulting in better solubility of hemicellulose as well as lignin. Notably, the DESs 1–3 had the best pretreatment efficiency, where cellulose retention, hemicellulose and lignin removal were 96%, 89%, and 74%, respectively, which might be because the active proton (Cl^-) in HBA could compete with the carbohydrates and lignin to form H-bond, also could break the ether bond in LCC (Lignin-Carbohydrate Complex) (Chen et al., 2022).

In addition, the carboxyl group in betaine hydrochloride could also form strong H-bond interactions. As shown in Table 4, the content of the by-products, such as formic acid, acetic acid, and furfural, was at a very low level, and the 5-HMF was not found in the pretreatment liquor. Compared to the glyceryl-DESs, the lactic acid-based DESs could accelerate the furfural generation, as verified by Lee et al. (2019). It demonstrated that the acidic carboxylic group in DESs significantly cracked the β-1,4 glycosidic bonds.

In accordance with the pretreatment results, the enzymatic efficiency of the samples that pretreated using DESs 1–3 was also at the highest level (92%), as shown in Fig. 1b. The enzymatic efficiency was positively correlated with the degradation of lignin and hemicellulose, which could expose more contact points between the cellulose and cellulase (Yu et al., 2016). Interestingly, though the pretreatment efficiency of the DESs 2–3 was not excellent, the enzymatic efficiency was still as high as 89%, indicating that the alkaline co-solvent could easily swell the fibrillar structure and reduce the interaction of the H-bond network. Furthermore, the cationization of glycerol and cellulose by betaine hydrochloride could enhance fibrillar swelling (Hong et al., 2020). At the same time, the hemicellulose removal of the sample pretreated using DESs 2–3 was obviously higher than that of other glycerol-DESs due to the action of betaine hydrochloride chloride ion.

3.3. Optimization of the pretreatment efficiency

The pretreatment conditions of the pretreatment using DESs 1–3 were further optimized using Box-Behnken design in response surface

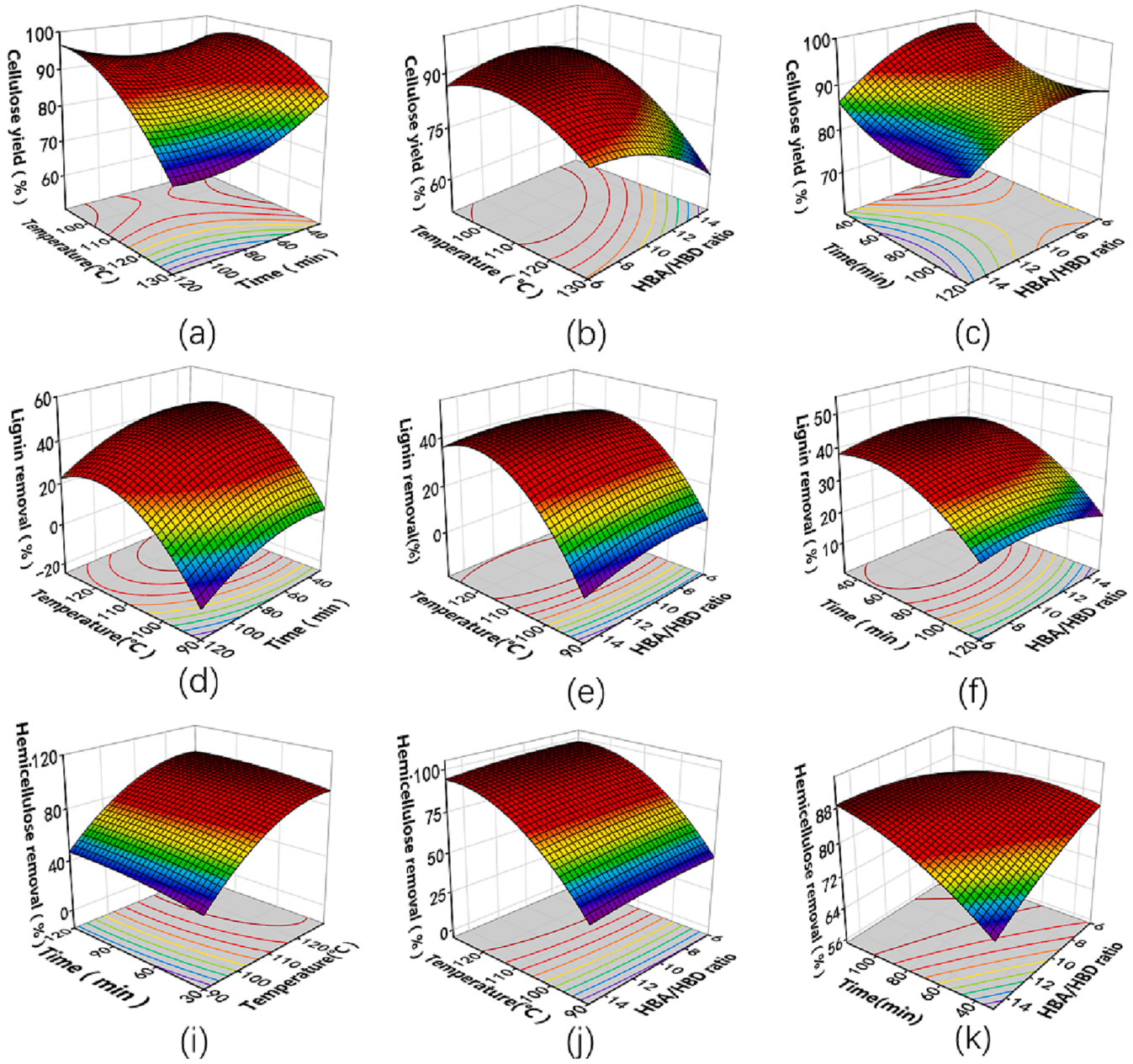


Fig. 2. Optimized the pretreatment conditions using Response surface methodology. Effects of the molar ratio, pretreatment temperature and time on the cellulose retention (a-c), the removal rates of lignin (d-f) and hemicellulose (i-k).

methodology, and the variation of the main components in the pretreated residues was shown in Table 2. The analysis of variance (ANOVA) showed (see supplementary material) that the *p*-values of their ANOVAs were all less than 0.05, indicating that the models were meaningful. Based on the above principles, it was shown that the model terms of A, B, C, AB, AC, A^2, B^2 and C^2 were considered significant for cellulose retention. According to the *F* value, the cellulose retention was influenced with the following orders: pretreatment temperature (A) > molar ratio of HBD to HBA (C) > pretreatment time (B). In Table 2, the cellulose retention of the samples (e.g., No. 5 and No. 6) was little changed with the pretreatment time extending under the conditions of fixed temperatures and a specific molar ratio. However, when pretreatment temperature came up to 130 °C, the cellulose retention of the samples (e.g., No.12 and No.13) was clearly decreased with the pretreatment time prolonging. The cellulose molecule with dense and order structures was formed by the intra- and intermolecular H-bonds, and with the lignin and hemicellulose as the protective layer on its surface, demonstrating that the decomposition of the cellulose only conducted under more intense conditions (Mankar et al., 2021a). As shown in Fig. 2 (d – k), the factors that affected the lignin and hemicellulose removal were in the following order: A > B > C. When the pretreatment temperature was raised from 90 °C to 110 °C, the removal rates of lignin and hemicellulose were increased by 37% and 40%, respectively. Meanwhile, it also expounded that the solution viscosity could be reduced with the temperature increasing (Procentese et al., 2015), making mass transfer more accessible, which might improve the pretreatment efficiency. In addition, according to the Arrhenius formula, the surface reaction kinetics could be greatly improved by the pretreatment temperature (Mamilla et al., 2019). Comparing to the lignin and cellulose removal, it was illustrated in Fig. 2(a, b, d, e, i and j) that the effects of the pretreatment temperature on the hemicellulose removal was much more significant than that of pretreatment time and molar ratio.

Based on the above analysis and the results from the response surface plots, the optimized pretreatment conditions were as follows:

Table 3
Kamlet–Taft parameters of three-constituent deep eutectic solvents.

No.	Metal chloride	Molar ratio	Kamlet–Taft parameters		
			π*	α	β
1	$AlCl_3$	14.5:1:0.1	1.00	2.08	0.56
2	$FeCl_3·6H_2O$		1.04	2.05	0.47
3	$MgCl_2$		1.02	2.05	0.52
4	$ZnCl_2$		1.04	2.05	0.44
5	$CuCl_2·2H_2O$		1.02	2.07	0.52
6	KCl		1.04	2.04	0.52
7	NaCl		1.04	2.04	0.44
8	NH_4Cl		1.04	2.04	0.52

Table 4
Concentration of the by-products in the pretreatment solution.

No.	Serial Number	FA (g/L)[a]	AA (g/L)[b]	5-HMF (g/L)[c]	FF (g/L)[d]
1	1–1	0.331 ±0.005	0.491 ±0.003	0.009 ±0.0005	0.073 ±0.003
2	1–2	0.165 ±0.002	0.375 ±0.001	N.D.	0.041 ±0.002
3	1–3	0.144 ±0.003	0.623 ±0.004	N.D.	0.061 ±0.003
4	1–4	0.051 ±0.001	0.431 ±0.003	N.D.	0.062 ±0.001
5	2–1	0.173 ±0.002	0.351 ±0.002	N.D.	0.023 ±0.003
6	2–2	0.182 ±0.003	0.336 ±0.003	N.D.	0.022 ±0.001
7	2–3	0.212 ±0.001	0.362 ±0.002	N.D.	0.031 ±0.004
8	2–4	0.156 ±0.004	0.305 ±0.005	N.D.	0.018 ±0.001

[a] FA stands for formic acid.
[b] AA stands for acetic acid.
[c] 5-HMF stands for 5 hydroxymethyl furfural.
[d] FF stands for furfural.

pretreatment temperature 105 °C, pretreatment time 145 min, and the molar ratio of DESs 14.5:1, where the theoretical removal rate of the cellulose, hemicellulose and lignin was 6%, 82% and 45%, respectively. In factually, the removal rate of the cellulose, hemicellulose and lignin came up to 6%, 92% and 63%, respectively. The predictions are basically consistent with the observation.

3.4. Polarity analysis of three-constituent deep eutectic solvent

For further improving the pretreatment efficiency and reducing the energy consumption, metal chloride salts were introduced into the two-consistent DESs, which could provide more acidic sites and had Lewis acid effects on the raw material pretreatment. Hence, eight types of the chloride salts containing KCl, NaCl, NHCl, $MgCl_2$, $ZnCl_2$, $CuCl_2$, $AlCl_3$ and $FeCl_3$ were utilized as the third components of the DESs to pretreat the lignocellulosic biomass.

The influence of the metal chloride salts on the DESs was determined by the Kamlet-Taft parameters. In Table 3, all of the 3c-DESs had higher α values, indicating that the 3c-DESs could easily donate H-bonds and had high acidity. In particular, the promotion effect of metal chloride on the α value for glycerol-based DESs was greater than that of lactic acid-based DESs. The α value was elevated from 0.77 to 1.99 with the $AlCl_3$ added to glycerol/choline chloride, which might be due to the metal chloride salts provided a large number of active protons could not only weaken the internal H-bond in glycerol, but also added more acidic sites. The H-bond sites had been sufficient enough in the lactic acid/betaine hydrochloride and challenging to form more acidic sites even added metal chloride salts, so the α value changed slightly. However, the β value fluctuated with the addition of metal salts, which was probably corresponded to the formation of Lewis acid that weakened the H-bond receptor. For the π^* value, it was did not changed with the addition of metal salts. Overall, the metal chloride salts added into the acidic DESs could also provide Lewis-acid effects in addition to the strong H-bond.

3.5. Effects of the three-constituent deep eutectic solvent on the pretreatment and enzymolysis efficiency

After adding metal chloride salts with different valence in DESs, the contents of hemicellulose and lignin in solid residues were further decreased, significantly varying with the types of metal ions in the chloride salts, as shown in In Fig. 3(a). The trivalent metal ion had a positive influence on the degradation of hemicellulose and lignin, while this influence was obviously weakened for the monovalent metal ion, indicating that the reduction of these two components was in direct proportion to the valence state of the metal ions. The addition of the metal chloride salts could provide the active protons (Cl^-) for the DESs, which competed with betaine hydrochloride and formed stronger H-

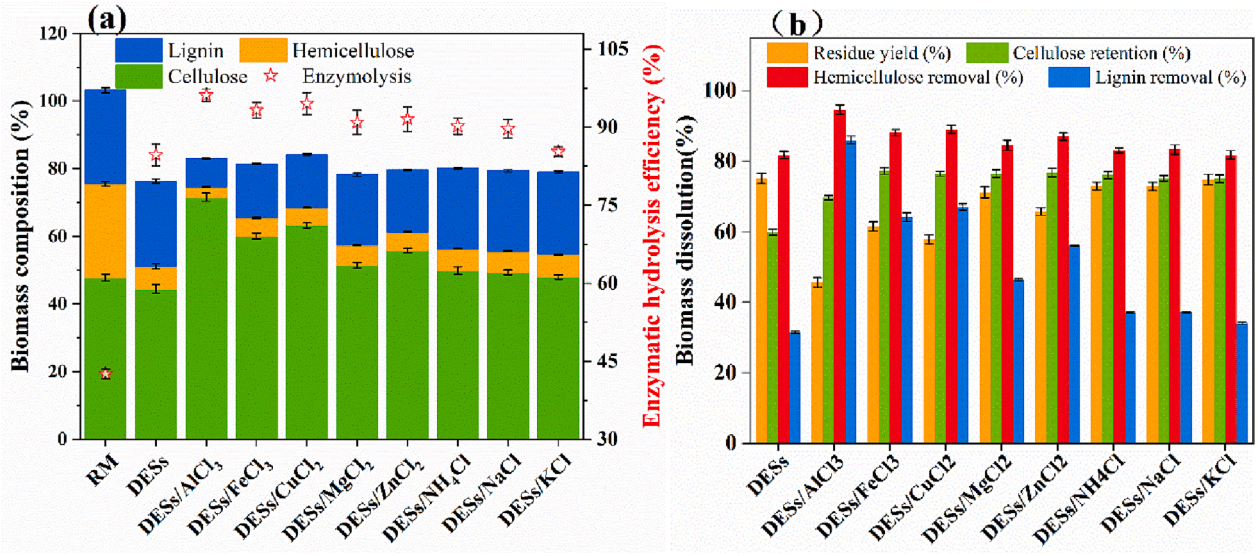


Fig. 3. Comparison of pretreatment efficiency for the three-constituent deep eutectic solvents: (a) Chemical components distribution and enzymatic efficiency of raw material (RM) and pretreatment residues. (b) Changes of the cellulose yield, and the removal rates of the hemicellulose and lignin in the pretreatment.

Table 5
Concentration distribution of byproducts in three-constituent deep eutectic solvents pretreatment solution.

No.	Pretreatment	FA [a] (g/L)	AA [b] (g/L)	5-HMF [c] (g/L)	FF [d] (g/L)
1	DESs/$FeCl_3$	0.642 ±0.005	0.713 ±0.003	0.034 ±0.001	0.123 ±0.002
2	DESs/$AlCl_3$	0.831 ±0.003	0.894 ±0.002	0.046 ±0.003	0.148 ±0.003
3	DESs/$MgCl_2$	0.633 ±0.004	0.683 ±0.003	0.029 ±0.002	0.116 ±0.001
4	DESs/$ZnCl_2$	0.671 ±0.002	0.721 ±0.004	0.031 ±0.003	0.129 ±0.005
5	DESs/$CuCl_2$	0.629 ±0.003	0.722 ±0.002	0.033 ±0.002	0.131 ±0.001
6	DESs/NaCl	0.381 ±0.002	0.692 ±0.003	0.022 ±0.001	0.107 ±0.003
7	DESs/KCl	0.368 ±0.005	0.653 ±0.004	0.025 ±0.003	0.112 ±0.002
8	DESs/NH_4Cl	0.411 ±0.002	0.629 ±0.005	0.019 ±0.002	0.098 ±0.004

[a] FA stands for formic acid.
[b] AA stands for acetic acid.
[c] 5-HMF stands for 5 hydroxymethyl furfural.
[d] FF stands for furfural.

bond to split the ether bonds in LCC (Xia et al., 2018). Therefore, the metal chloride salts with high valence state had better H-bond donor ability, which could be illustrated by their α value changes (Table 3). At the same time, because of the three-electron redox properties of the $AlCl_3$, electron-rich oxygen could be attracted to form a polyhedral coordination cage centered on the Al^{3+}, and the formation of specific H-bonds with lactic acid further enhanced the proton transfer ability of the lactic acid, which promoted the destruction of the H-bond network of the lignocellulosic biomass. In the Fig. 3(b) concerning on the components variation, the pretreatment using DESs/$AlCl_3$ had the best selective fractionation efficiency, where the removal rates of the hemicellulose and lignin were as high as 95% and 86%, respectively.

In accordance with the pretreatment efficiency, the generation of by-products was also positively correlated with the valence state of the metal ions. When the 3c-DESs were used in the pretreatment instead of the two-constituent DESs, the concentration of all the by-products was increased (Table 5), meaning that the removal rates of hemicellulose and lignin were promoted by the Lewis acidic effects of the metal chloride salts. However, the further degradation of monosaccharides was also intensified that the contents of FF, 5-HMF, small molecular acids etc. were distinctly raised, which was not conducive to the DESs recycling the high-value utilization of sugars in the pretreatment solution (Chen et al., 2022).

It was clearly shown in Fig. 3(a) that the enzymolysis efficiency was significantly improved by pretreatment using DESs, which was further enhanced by the metal chloride salts in the DESs, with a positive relationship to the valence state of the metal ions. Along with the hemicellulose and lignin degradation in the pretreatment, more attachment sites between the cellulose and cellulase were exposed, improving the enzymolysis efficiency. Therefore, the highest enzymolysis efficiency was emerged in the pretreatment using DESs-$AlCl_3$, as high as 96%.

3.6. Molecular structure changes for the raw material and solid residues

In the Fig. 4(a), the XRD results showed that typical diffraction peaks associated with cellulose *I* ($2\theta = 14.5°$, 16.5° and 22.5°) were found in the raw material and all of the pretreated solid residues, indicating that the pretreated cellulose still retained its original structure. Comparing to the raw material, the *CrI* of the pretreated solid residues was obviously increased. In particular, the *CrI* of the solid residues from the pretreatment using DESs-$AlCl_3$ was raised about 9%, this value was clearly lower than that of using the other 3c-DESs (about 18%). Compared with other metal chloride salts, the $AlCl_3$ could provide more active sites (Cl^-), and the Al^{3+} with high activity could form polyhedral coordination species to enhance the ability of donor H-bond. As a result, the glycosidic bonds in the crystalline region of cellulose were also destroyed. Thus, the *CrI* of the solid residues was increased at a low level (Liu et al., 2016).

The molecular changes of the raw material in the pretreatment are illustrated in Fig. 4(b), and the assignment of the absorption peaks has been reported in detail in the references (Sai and Lee, 2019). The typical absorption peaks at 1594, 1504, 1461 and 1423 cm^{-1} were assigned to the aromatic ring in lignin, their signal intensity was obviously weakened after the pretreatment using DESs and 3c-DESs. The peaks at 1594 and 1504 cm^{-1} were the skeleton vibrations of the aromatic ring, where the peak at 1461 cm^{-1} was the signal of C-H asymmetric deformation, and the peak for the C-H deformation in the plane of the skeleton vibrations of the aromatic ring was found at 1423 cm^{-1}. The most variation of the molecular structural for the solid residues was emerged in the pretreatment using DESs/$AlCl_3$, DESs/$FeCl_3$, and DESs/$CuCl_2$, indicating that the delignification effect of metal ions with high valence state was more efficient.

The absorption peaks concerning on the cellulose were emerged at 1373, 1326, 1162, 1040 and 898 cm^{-1}, where 1373, 1326 and 898 cm^{-1} represented the characteristic peaks of type I cellulose, and no signal changes was found after the pretreatment using 3c-DESs, this result was in accordance with the XRD analysis. The characteristic absorption peaks at 1040 and 898 cm^{-1} represented the glycosidic bonds of cellulose, and the signal intensity was not improved with the pretreatment using 3c-DESs, namely the cellulosic glycoside bond was very stable (Ha et al., 2011). In addition, the peak at 1735 cm^{-1} was assigned to the C = O stretching vibration, the enhancement of the signal corresponding to the esterification reaction between the hydroxyl groups in cellulose and the lactic acid, which enhanced the repulsive force among the fibers and formed a stable, negatively charged suspension (Fig. 4(b)).

3.7. Microstructure characterization of the raw material and solid residues

Microstructural changes of the raw material during the pretreatment are plotted (see supplementary material). It was evident that most of the fiber bundles of the raw material were tightly combined and wrapped by rough lamellar structures (see supplementary material), which hindered the accessibility of the cellulase to cellulose. After pretreatment, the connection among the single-fiber or fiber bundle was loosed, producing more cracks and pores. In particular, the fibrous bundle from the pretreatment using DESs/$AlCl_3$ (see supplementary material) was seriously damaged, with a transverse fracture on the cross-sectional surface. Ultimately, the perimeter protection "barriers" for the cellulose was removed, along with a small proportion of the cellulose dissolved.

4. Conclusions

The lactic acid-based DESs with high polarity had significant efficiency for the pretreatment and enzymolysis. The carboxyl group, Cl^- and N^+ could promote the breakage of glycosidic-, ether- and hydrogen-bonds. The cellulose was partially esterified and improved the enzymolysis. The DESs combined with trivalent metal chlorides had abundant active sites and further depolymerized the hemicellulose and lignin, which formed porous microstructures in solid residues. The DESs-$AlCl_3$ had the best pretreatment efficiency, where the removal rates of lignin and hemicellulose, and enzymolysis efficiency were 86%, 95% and 96%, respectively. It provided a new perspective for efficiently separating lignocellulose from green solvents.

CRediT authorship contribution statement

Dan Huo: Conceptualization, Funding acquisition, Investigation, Writing – original draft, Writing – review & editing. **Yuekai Sun:**

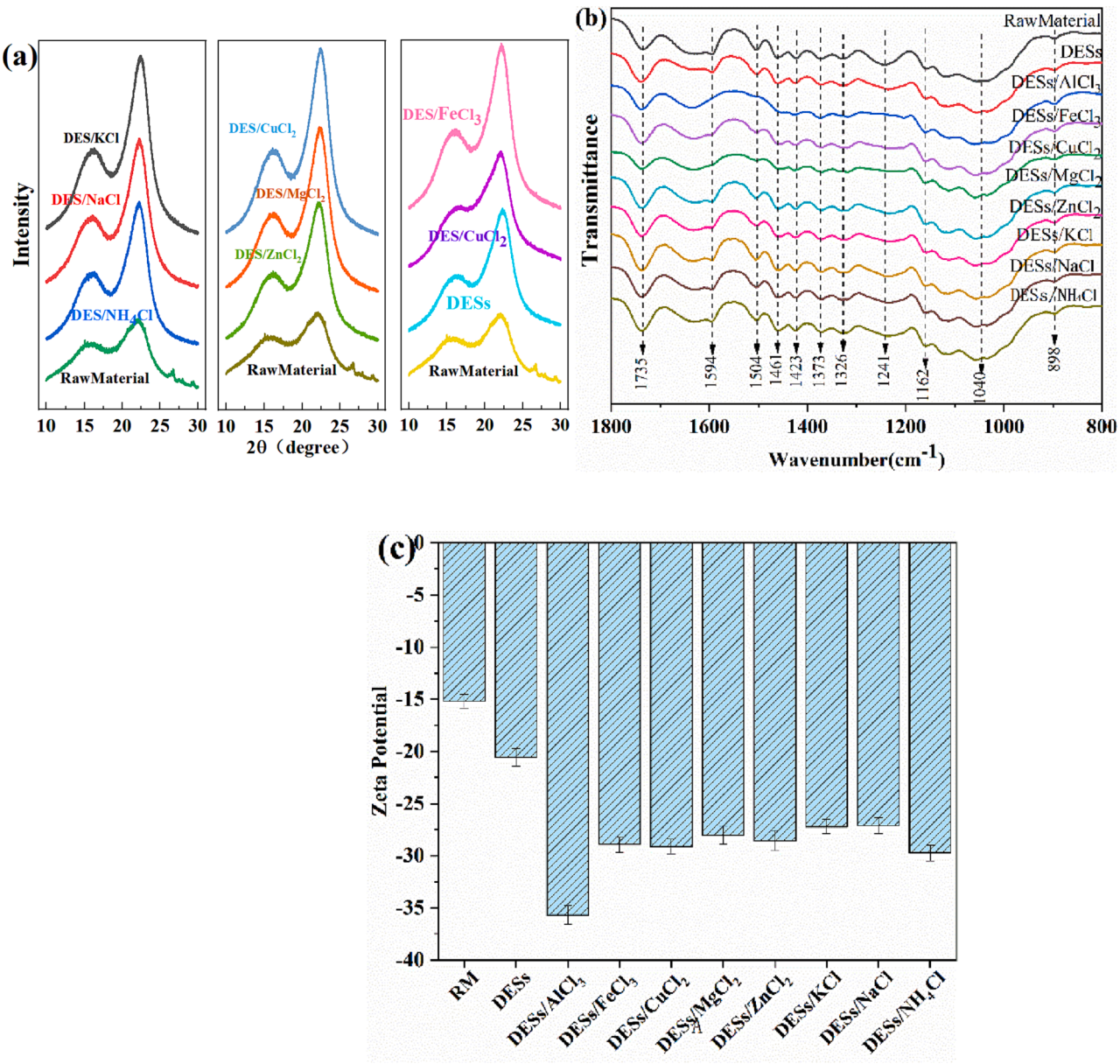


Fig. 4. Characteristics of the raw materials (RM) and pretreatment residues: (a) Crystallinity index (b) Fourier transform infrared spectroscopy (c) Zeta potential analysis.

Investigation, Formal analysis, Writing – original draft. **Qiulin Yang:** Investigation, Resources, Writing – review & editing. **Fengshan Zhang:** Conceptualization, Formal analysis. **Guigan Fang:** Methodology, Investigation, Writing – review & editing. **Hongxiang Zhu:** . **Ying Liu:** Methodology, Investigation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biortech.2023.128937.

References

Abbott, A.P., Boothby, D., Capper, G., Davies, D.L., Rasheed, R.K., 2004. Deep eutectic solvents formed between choline chloride and carboxylic acids: Versatile alternatives to ionic liquids. J Am Chem Soc 126, 9142–9147.

Carriazo, D., Serrano, M.C., Gutiérrez, M.C., Ferrer, M.L., del Monte, F., 2012. Deep-eutectic solvents playing multiple roles in the synthesis of polymers and related materials. Chem Soc Rev 41, 4996–5014.
Chen, Z., Bai, X.L., Lusi, A., Wan, C.X., 2018. High-Solid Lignocellulose Processing Enabled by Natural Deep Eutectic Solvent for Lignin Extraction and Industrially Relevant Production of Renewable Chemicals. ACS Sustainable Chem. Eng. 6, 12205–12216.
Chen, Z., Wang, Y., Cheng, H., Zhou, H., 2022. Hemicellulose degradation: An overlooked issue in acidic deep eutectic solvents pretreatment of lignocellulosic biomass. Ind. Crops Prod. 187, 115335.
Florindo, C., McIntosh, A.J.S., Welton, T., Branco, L.C., Marrucho, I.M., 2018. A closer look into deep eutectic solvents: exploring intermolecular interactions using solvatochromic probes. Phys Chem Chem Phys 20, 206–213.
Ha, S. H.; Mai, N. L.; An, G.; Koo, Y.-M. J. B. t. Microwave-assisted pretreatment of cellulose in ionic liquid for accelerated enzymatic hydrolysis. Bioresour. Technol. 2011. 102. 1214-1219.
Hong, S., Yuan, Y., Li, P., Zhang, K., Liimatainen, H., 2020. Enhancement of the nanofibrillation of birch cellulose pretreated with natural deep eutectic solvent[J]. Ind. Crops Prod. 154, 112677.
Hou, X.D., Li, A.L., Lin, K.P., Wang, Y.Y., Kuang, Z.Y., Cao, S.L., 2018. Insight into the structure-function relationships of deep eutectic solvents during rice straw pretreatment. Bioresour. Technol. 249, 261–267.
Kamireddy, S.R., Li, J.B., Tucker, M., Degenstein, J., Ji, Y., 2013. Effects and Mechanism of Metal Chloride Salts on Pretreatment and Enzymatic Digestibility of Corn Stover. Ind Eng Chem Res 52, 1775–1782.
Kandanelli, R., Thulluri, C., Mangala, R., Rao, P.V.C., Gandham, S., Velankar, H.R., 2018. A novel ternary combination of deep eutectic solvent-alcohol (DES-OL) system for synergistic and efficient delignification of biomass. Bioresour. Technol. 265, 573–576.
Kim, J.K., Yang, J., Park, S.Y., Yu, J.-H., Kim, K.H., 2019. Cellulase recycling in high-solids enzymatic hydrolysis of pretreated empty fruit bunches. Biotechnol. Biofuels 12, 138–147.
Kumar, A.K., Parikh, B.S., Pravakar, M., 2016. Natural deep eutectic solvent mediated pretreatment of rice straw: bioanalytical characterization of lignin extract and enzymatic hydrolysis of pretreated biomass residue. Environ. Sci. Pollut. Res. 23, 9265–9275.
Lee, S.H., Doherty, T.V., Linhardt, R.J., Dordick, J.S., 2009. Ionic Liquid-Mediated Selective Extraction of Lignin From Wood Leading to Enhanced Enzymatic Cellulose Hydrolysis. Biotechnol. Bioeng. 102, 1368–1376.
Lee, C.B.T.L., Wu, T.Y., Ting, C.H., Tan, J.K., Siow, L.F., Cheng, C.K., Jahim, J.M., Mohammad, A.W., 2019. One-pot furfural production using choline chloride-dicarboxylic acid based deep eutectic solvents under mild conditions. Bioresour. Technol. 278, 486–489.
Li, W.Q., Amos, K., Li, M., Pu, Y.Q., Debolt, S., Ragauskas, A.J., Shi, J., 2018. Fractionation and characterization of lignin streams from unique high-lignin content endocarp feedstocks. Biotechnol. Biofuels 11, 1–14.
Li, T.F., Lyu, G.J., Liu, Y., Lou, R., Lucia, L.A., Yang, G.H., Chen, J.C., Saeed, H.A.M., 2017. Deep Eutectic Solvents (DESs) for the Isolation of Willow Lignin (Salix matsudana cv. Zhuliu). Int. J. Mol. Sci. 18, 2266–2277.
Liu, C., Li, B., Du, H.S., Lv, D., Zhang, Y.D., Yu, G., Mu, X.D., Peng, H., 2016. Properties of nanocellulose isolated from corncob residue using sulfuric acid, formic acid, oxidative and mechanical methods. Carbohydr. Polym. 151, 716–724.
Ma, J., Ji, Z., Chen, J.C., Zhou, X., Kim, Y.S., Xu, F., 2015. The mechanism of xylans removal during hydrothermal pretreatment of poplar fibers investigated by immunogold labeling. Planta 242, 327–337.
Mamilla, J.L.K., Novak, U., Grilc, M., Likozar, B., 2019. Natural deep eutectic solvents (DES) for fractionation of waste lignocellulosic biomass and its cascade conversion to value-added bio-based chemicals. Biomass Bioenergy 120, 417–425.
Mankar, A.R., Pandey, A., Modak, A., Pant, K.K., 2021a. Pretreatment of lignocellulosic biomass: A review on recent advances. Bioresour. Technol. 334, 125235–125247.
Mankar, A.R., Pandey, A., Pant, K.K., 2021b. Microwave-assisted extraction of lignin from coconut coir using deep eutectic solvents and its valorization to aromatics. Bioresour. Technol. 345, 126528–126537.
Pandey, A., Rai, R., Pal, M., Pandey, S., 2014. How polar are choline chloride-based deep eutectic solvents? Phys Chem Chem Phys 16, 1559–1568.
Procentese, A., Johnson, E., Orr, V., Campanile, A.G., Wood, J.A., Marzocchella, A., Rehmann, L., 2015. Deep eutectic solvent pretreatment and subsequent saccharification of corncob. Bioresour. Technol. 192, 31–36.
Qiu, J., Wang, Q., Shen, F., Yang, G., Zhang, Y., Deng, S., Zhang, J., Zeng, Y., Song, C., 2017. Optimizing phosphoric acid plus hydrogen peroxide (PHP) pretreatment on wheat straw by response surface method for enzymatic saccharification. Appl. Biochem. Biotechnol. 181, 1123–1139.
Sai, Y.W., Lee, K.M., 2019. Enhanced cellulase accessibility using acid-based deep eutectic solvent in pretreatment of empty fruit bunches. Cellulose 26, 9517–9528.
Wang, H., Li, J., Zeng, X., Tang, X., Sun, Y., Lei, T., Lin, L., 2020. Extraction of cellulose nanocrystals using a recyclable deep eutectic solvent. Cellulose 27, 1301–1314.
Wang, J., Wang, P., Wang, Q., Mou, H., Cao, B., Yu, D., Wang, D., Zhang, S., Mu, T., 2018. Low Temperature Electrochemical Deposition of Aluminum in Organic Bases/Thiourea-Based Deep Eutectic Solvents. ACS Sustainable Chem. Eng. 6, 15480–15486.
Xia, Q.Q., Liu, Y.Z., Meng, J., Cheng, W.K., Chen, W.S., Liu, S.X., Liu, Y.X., Li, J., Yu, H. P., 2018. Multiple hydrogen bond coordination in three-constituent deep eutectic solvents enhances lignin fractionation from biomass. Green Chem 20, 2711–2721.
Yiin, C.L., Yap, K.L., Ku, A.Z.E., Chin, B.L.F., Lock, S.S.M., Cheah, K.W., Loy, A.C.M., Chan, Y.H., 2021. Recent advances in green solvents for lignocellulosic biomass pretreatment: Potential of choline chloride (ChCl) based solvents. Bioresour. Technol. 333, 125195–125207.
Yu, Q., Liu, J., Zhuang, X.S., Yuan, Z.H., Wang, W., Qi, W., Wang, Q., Tan, X.S., Kong, X. Y., 2016. Liquid hot water pretreatment of energy grasses and its influence of physico-chemical changes on enzymatic digestibility. Bioresour. Technol. 199, 265–270.
Zhang, C.W., Xia, S.Q., Ma, P.S., 2016. Facile pretreatment of lignocellulosic biomass using deep eutectic solvents. Bioresour. Technol. 219, 1–5.